# The t Test for One or Two Means

Lectures 22–23

QBA201 – Quantitative Business Analysis

Shi Qiu – Fall 2026

# What changes when we estimate the spread?

## Recall the denominator

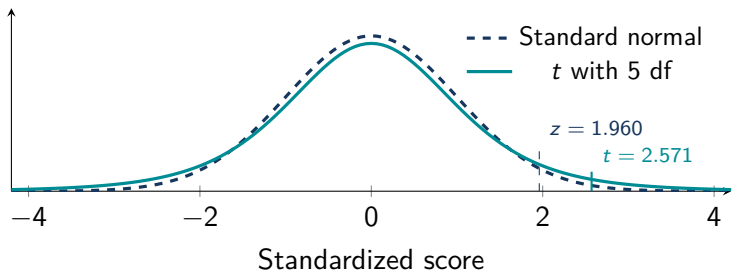
For a mean test, the numerator is the observed mean minus the benchmark: $\bar{x} - \mu_0$.

| Population SD | Standard error used | Reference under $H_0$ |
|---|---|---|
| Known $\sigma$ | $\sigma/\sqrt{n}$ | Standard normal $Z$ |
| Unknown $\sigma$ | $s/\sqrt{n}$ | Student $t$ with $n-1$ degrees of freedom |

For independent normal observations, $\bar{X}$ stays normal. The statistic with the estimated standard error follows $t$.

# A small sample needs a wider reference

If $n = 6$, the $t$ reference has five degrees of freedom. Compare the upper cutoffs leaving 2.5% on their own curves.




The $t$ tails are heavier. As degrees of freedom increase, the curve and its cutoffs approach the standard normal.

# Build a t score for an average cost

An independent random sample of 25 normal parcel-handling costs gives mean AED 18.80 and sample SD AED 2.00. Test whether the population mean differs from AED 18. Use $\alpha = .05$.

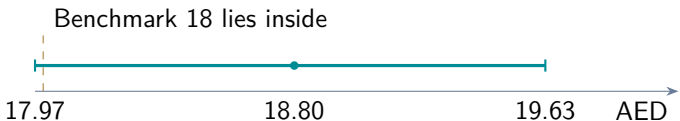
$$H_0 : \mu = 18, \qquad H_1 : \mu \neq 18.$$

## Calculate the gap in estimated standard errors

$$\widehat{\mathrm{SE}} = \frac{2}{\sqrt{25}} = 0.40, \qquad t_{\mathrm{obs}} = \frac{18.80 - 18}{0.40} = 2.00, \qquad \nu = 24.$$

# Check the decision against the interval

For the two-sided question, $p = 2\,\mathrm{P}(T_{24} \geq 2) = 0.0569$. At 5%, fail to reject $H_0$.




95% interval : $\quad 18.80 \pm 2.064(0.40) \approx [17.97, 19.63]$.

The interval includes AED 18. The data leave the difference unresolved at 5%.

# Use the signed t score for a directional claim

With 24 degrees of freedom, consider three separate preplanned questions for $t_{\text{obs}} = 2.00$.

| Alternative | Area under $t_{24}$ | p-value | At 5% |
|---|---|---|---|
| $\mu > \mu_0$ | Right of 2.00 | .0285 | Reject |
| $\mu < \mu_0$ | Left of 2.00 | .9715 | Fail to reject |
| $\mu \neq \mu_0$ | Both beyond $\pm 2.00$ | .0569 | Fail to reject |

A negative score of the same size reverses the one-tail areas. Use the alternative chosen before sampling.

# Check whether the sampling design changes the SE

## Independent normal observations

For an unknown population SD, $s/\sqrt{n}$ is the denominator for the exact one-sample $t$ result.

## Sample without replacement from a fixed list

For a simple random sample of $n$ from a list of $N$ units, a design-based estimated SE is

$$\frac{s}{\sqrt{n}}\sqrt{1-\frac{n}{N}}.$$

The adjustment reduces estimated variability. The independent-normal $t$ reference needs its own assumptions.

# Try a left-tail t test for delivery time

A logistics firm wants to know whether its mean delivery time is below 45 minutes. For an independent random sample of 16 delivery times from a normal population, $\bar{x} = 42.5$ and $s = 5.0$ minutes. Use $\alpha = .05$.

## Choose the tail and the reference

$$H_0 : \mu \geq 45, \qquad H_1 : \mu < 45, \qquad t_{\text{obs}} = \frac{42.5 - 45}{5/\sqrt{16}} = -2.00, \qquad \nu = 15.$$

## Check the business decision

The left-tail area is $\mathrm{P}(T_{15} \leq -2.00) = .0320$. Reject at 5%. The sample supports a population mean delivery time below 45 minutes.

# From one mean to a difference of means

Let $\mu_A$ and $\mu_B$ be the population means of two independent groups. The observed difference is $\bar{x}_A - \bar{x}_B$. For an equality claim, the null difference is zero.

| What we know about the spread | Reference | Standard error uses |
|---|---|---|
| Both population SDs known | $z$ | Known population SDs |
| Unknown SDs, no common spread assumed | Welch $t$ | Two separate sample variances |
| Unknown SDs, common spread justified | Pooled $t$ | One pooled variance estimate |

Decide whether a common variance is justified from the setting. The observed p-values do not choose the procedure.

# Set up a two-sample Welch t test

Two suppliers use different processes. A buyer checks whether their mean handling costs differ. Each sample is independent and random, and costs in each population are normal. The population variances are unknown and need not be equal.

| Supplier | Sample size $n$ | Sample mean (AED) | Sample SD (AED) |
|---|---|---|---|
| A | 16 | 42.50 | 4.00 |
| B | 25 | 39.00 | 8.00 |

$$H_0 : \mu_A - \mu_B = 0, \qquad H_1 : \mu_A - \mu_B \neq 0.$$

The estimated gap is AED 3.50. We need to measure its uncertainty using both samples.

# Estimate each group's contribution to uncertainty

Let $a = s_A^2/n_A$ and $b = s_B^2/n_B$ be the estimated sampling variances of the two means.

$$a = \frac{4^2}{16} = 1.00, \qquad b = \frac{8^2}{25} = 2.56.$$

**Add variances, then take the square root**

$$\widehat{\mathrm{SE}} = \sqrt{a+b} = \sqrt{3.56} \approx 1.8868, \qquad t_{\mathrm{obs}} = \frac{(42.50 - 39.00) - 0}{\sqrt{3.56}} \approx 1.8550.$$

**Find the reference degrees of freedom**

$$\nu \approx \frac{(a+b)^2}{a^2/(n_A - 1) + b^2/(n_B - 1)} = \frac{3.56^2}{1.00^2/15 + 2.56^2/24} \approx 37.3046.$$

# Make the decision using the Welch reference

For the planned two-sided question, $t_{\text{obs}} \approx 1.8550$ with approximately 37.3046 degrees of freedom.

| Planned alternative | Tail area | Decision at 5% |
|---|---|---|
| $\mu_A \neq \mu_B$ | Both, .0715 | Fail to reject |
| $\mu_A > \mu_B$ | Right, .0358 | Reject |
| $\mu_A < \mu_B$ | Left, .9642 | Fail to reject |

For the buyer's two-sided question, the evidence falls short at 5%. The other rows answer different questions chosen in advance.

# Pool variances when a common spread is justified

Two repair depots use the same process and equipment. Assume independent random samples from normal populations with a common unknown variance. Test whether their population mean repair times differ.

| Depot | $n$ | Sample mean (minutes) | Sample SD (minutes) |
|---|---|---|---|
| A | 12 | 31 | 4 |
| B | 10 | 26 | 5 |

## Estimate the common variance

$$s_p^2 = \frac{(n_A - 1)s_A^2 + (n_B - 1)s_B^2}{n_A + n_B - 2} = \frac{11(16) + 9(25)}{20} = 20.05.$$

# Make the decision using the pooled reference

For $\mu_A$ and $\mu_B$ the population mean repair times, test $H_0 : \mu_A - \mu_B = 0$ against $H_1 : \mu_A - \mu_B \neq 0$.

**Use the common estimate in both sampling variances**

$$\widehat{\mathrm{SE}} = \sqrt{20.05(1/12 + 1/10)} \approx 1.9172, \qquad t_{\mathrm{obs}} = \frac{(31 - 26) - 0}{\widehat{\mathrm{SE}}} \approx 2.6079.$$

**Use $n_A + n_B - 2 = 20$ degrees of freedom**

Two-sided $p \approx .0168$. Reject at 5%. The data support different population mean repair times.

A common variance is an assumption about the two populations. It is compatible with different means.